Fiscal Year:	FY 2012	**Task Last Updated:**	FY 05/21/2012
PI Name:	Poirier, David Sc.D.		
Project Title:	Research for ESA Space Experiments on Directional Solidification of Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721-0012	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	05/15/2008	**End Date:**	05/14/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	3
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS Expedition 19/20 NOTE: Extended to 5/14/2014 per PI and NSSC information (Ed., 4/5/2013) NOTE: End date extended to 5/14/2013 (from 5/14/2011) in Dec 2010 per NSSC information (Ed., 2/14/2011)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Erdmann, Robert (University of Arizona) Tewari, Surendra (Cleveland State University)		
Grant/Contract No.:	NNX08AN49G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This is a program to conduct experiments in the International Space Station (ISS) to study the effect of microgravity on directionally solidified alloy microstructures and segregation. At Cleveland State University (CSU) terrestrial directional solidification experiments on Al-7%Si alloys have been done using different growth conditions to effect variations in dendritic metrics. Numerical modeling to simulate the transport phenomena and solidification of the planned experiments is currently underway at the University of Arizona (UA). Based on previous research, it is clear that thermosolutal convection masks the effects of diffusional transport during solidification and also leads to macro-segregates known as freckles. We believe, therefore, that with terrestrial experiments, diffusive and convective phenomena occur simultaneously, which complicate the study of dendritic growth. In microgravity, however, it would be possible to effect directional solidification with no thermosolutal convection. Our hypotheses are the convection and attendant macrosegregation will be eliminated and the dendritic microstructure will be greatly altered in microgravity. Availability of microgravity provides an opportunity to obtain experimental data, where thermosolutal convection is negligible. With this grant, the effect of solidifying in space versus solidifying on earth in the presence of gravity are studied by combining mathematical modeling and simulations with the empirical results of both terrestrial and space experiments. Thermosolutal convection will be significantly reduced in microgravity, and thus we can compare segregation and microstructure in specimens terrestrially solidified and those solidified in microgravity. The technological significance of the research is that it will provide scientific underpinning to our understanding and control of how single-crystal castings solidify. Single-crystal castings are critical components in high-temperature gas-turbine engines that are used in high-speed aircraft and land-based power generators. Currently we are doing research on the effect of convection on the microstructure and macrosegregation of directionally solidified castings. We are part of MICAST (Microstructure Formation in Casting of Technical Alloys) led by ESA-researchers in the German Aerospace Center in Germany. One sample was processed in the ISS in August, 2010 and a second was processed on the ISS in Feb. 20011. Both samples have been returned to Earth and have been under study at CSU and MSFC. Computer simulations have been ongoing at UA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is relevant to a process known as directional solidification, which is used for producing critical castings that are used in gas turbine engines. Gas turbine engines power aircraft (i.e., often called "jet engines") and are also used to generate electric power. These castings are made of vacuum melted alloys that are capable of withstanding the demanding conditions at high temperatures and consequently the metallurgical structure of the alloys must be of the highest quality. The research will lead to an improved understanding of how microstructures and segregation form in directionally solidified castings; hence the science of producing such castings will be enhanced as a result of the research. In addition, the research-findings will benefit many other types of castings, including billets and ingots, of alloys (i.e., steel, aluminum alloys, etc.) that are important in the "manufacturing chain."
Task Progress:	Two samples of Al-7%Si alloy were processed by directional solidification (DS) on the ISS. One was processed in August, 2010 and the second in Feb. 2011; both samples have been returned to Earth. Samples from counterpart terrestrial-DS experiments on the same alloy are being studied. One aspect of microstructure is the crystallographic orientation of the dendrites that result from the DS of the subject alloy. In terrestrial-DS samples the dendrites appear to have crystal orientations when viewed using a commonly employed microscopy technique. Using a more sophisticated technique (called Electron Back Scattered Diffraction), features of the dendrites (called tertiary dendrite arms) were found to be detached with crystal-orientations that differed from their dendrites. These arms can be considered to be defects in the regularity of the crystal structure of the alloy, so this is an important finding. Preliminary examinations, as of this writing, show that the same dendritic-defect is also found in the space-processed alloy. Reducing the gravity to almost zero in the ISS (and hence liquid motions) has little effect on preventing this particular defect. Other aspects of the dendrites resulting from DS are being examined. In particular, solidification in the presence of gravity (i.e., DS on Earth) causes severe motion of the liquid-phase, and this results in insidious defects known as freckles. When DS-ed in the micro-gravity environment of the ISS, however, the samples should be free of "freckles" and other related defects attributed to the liquid motion. Examination and measurements of the distribution of the alloy-element in specimens terrestrially solidified and those solidified in the ISS microgravity are still in progress. Since the liquid motion during solidification is rather complex, we are also performing computer simulations of the DS to assist in interpreting the results of our experiments. To effect such simulations, many physical properties of the subject alloy have been researched by collecting and assessing data from the technical literature. A computer program to simulate changes during the "hold-period" on the ISS has been completed and a development of a "sister" program to simulate the changes in the presence of the liquid motions (i.e., on Earth) is underway. Computer programs for simulating the DS-processing, both on Earth and in the ISS, have also been done.
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Articles in Peer-reviewed Journals	Van Hoose JR, Grugel RN, Tewari SN, Brush LN, Erdmann RG, Poirier DR. "Observation of mis-oriented tertiary arms during controlled directional solidification in aluminum-7wt% silicon alloys." Metallurgical and Materials Transactions B 2012; in press as of May 2012. , May-2012